

Project Proposal: Mobile-First Crop Disease Detection System

Khoa Viet Anh Nguyen, Thanh Chi Nguyen

Saigon International University

I. Introduction

Background and Scope

Our current global population is expected to exceed 9 billion by 2050, requiring a 70% increase in food production. However, agricultural output is currently greatly affected by infectious diseases, which drastically reduce potential yields by an average of 40%. Furthermore, in many developing regions, these losses can reach as high as 100%, damaging local economies and food security.

This project falls within the scope of Precision Agriculture and Mobile Health for Plants (mH-Plant). Our focus is on the development of an offline-capable smartphone application that allows farmers to identify crop pathologies in real-time by analyzing leaf imagery.

Problem Statement and Motivations

- **The Expertise Gap:** Developing regions often have a 1:5,000 ratio of agricultural experts to farmers (Africa Farm, 2024). This creates a “diagnostic bottleneck” where infections spread long before an expert can arrive. This results in farmers without any expertise guidance must resort to “blanket spraying” or using expensive, toxic, and in many cases ineffective chemicals, which harms both the environment and their limited income.
- **Visibility Problem:** Satellites, drones, and stationary cameras often provide a “macro view” that misses most of the early-stage symptoms. These systems do not have the capability to see under the leaf canopy or detect tiny (5mm) necrotic spots where infections begin. We have come up with a solution to address this issue, which is to use a handheld phone that provides macro-photography resolution. It should be able to capture the specific textures, pustule shape, and underside features required for an accurate diagnosis that a regular camera system cannot see.

- **Accessibility Problem:** Stationary IoT sensors and drones require high-bandwidth WiFi and thousands of dollars in infrastructure that usually exceed the annual income of smallholder farmers. This project utilizes the smartphone that most farmers already own. By optimizing models for on-device (TensorFlow Lite), we provide advanced diagnostics that function offline in remote fields without additional hardware costs.
- **Goal:** By providing a mobile AI tool, the farmer no longer has to wait days or weeks for an expert or government official to visit their remote fields. They will have the ability to make immediate, expert-level decisions on-site.

II. Related Works

2.1. Using Deep Learning for Image-Based Plant Disease Detection (Mohanty, Hughes, & Salathé, 2016)

This seminal study utilized a dataset of 54,306 images to train Deep Convolutional Neural Networks (CNNs). They demonstrated that models like AlexNet and GoogLeNet could achieve a classification accuracy of 99.35% in identifying 14 crop species and 26 diseases. While Mohanty's work proved the feasibility of AI-based diagnostics, the research highlighted a critical limitation: accuracy dropped significantly when models were tested on images taken under different lighting or background conditions. This established the need for segmented data to help models generalize beyond lab conditions.

2.2. Evolution of Mobile-Friendly Architectures

Traditional models like ResNet-50 are too computationally heavy for regular smartphone hardware. Research into MobileNetV3 (Howard, et al., 2019) and EfficientNet-Lite (Sangar & Rajasekar, 2025) introduced depthwise separable convolutions and inverted residuals. These lightweight architectures can maintain high accuracy while reducing the model size to under 30MB. This shows that high-performance AI no longer requires a powerful server and can be compressed for a farmer's handheld device.

2.3. Connectivity Barrier

Cloud-dependent AI is often unusable in rural agricultural areas due to high latency and poor 4G/5G coverage. Running the AI directly on the phone's CPU/GPU (Edge Inference) is our current solution for this issue. By processing images locally, these systems ensure data privacy and offline reliability, allowing for instant diagnosis in remote fields.

III. Project Outline

3.1. The Idea and Project Scope

This project aims to build a lightweight mobile application that allow user to point their phone at a leaf and receive instant diagnosis, confidence score, and treatment suggestions without an internet connection.

3.2. Abstract

Using the PlantVillage dataset, we will train a multi-class classifier. The key innovation lies in utilizing segmented images for training to ensure the model learns biological features (spots, wilting, color, etc.) rather than background noise (dirt, hands, sky). The final model will be converted to a mobile-optimized format (TFLite) for integration into a cross-platform mobile app.

3.3. Procedures

1. **Data Curation:** Filter more than 50,000 images from PlantVillage dataset. Split them into training, validation, and testing sets across 38 classes.
2. **Preprocessing and Augmentation:** Use OpenCV to resize images to 224x224. Apply albumentations (rotation, flip, brightness) to simulate the variability of outdoor smartphone photography.
3. **Model Training:** Implement transfer learning using a pre-trained EfficientNet-B0 or MobileNetV3 backbone.
4. **Optimization:** Use post-training quantization to reduce the model size to <10MB without significant loss in accuracy.
5. **Application Development:** Implement a user-friendly UI that captures images from the camera and runs them through the local TFLite interpreter.
6. **Offline Testing:** Verify that the app provides accurate results in airplane mode to simulate remote field conditions.

IV. Expected Outcome

Demo

A prototype mobile application where users can:

1. Take a photo of any corn leaf.
2. See instant label (e.g., “Corn Common Rust – 94% Confidence”).
3. Read a short treatment suggestion snippet.

Evaluation Methods

- Accuracy/F1-Score: Ensure balanced performance across all 38 classes.
- Model Size (MB): Ensure the app is small enough for low-storage devices.
- Confusion Matrix: Identify which diseases are visually similar and require further training/segmentation focus.